

9.2 Extra sums of squares

Most questions asked of a regression compare two models. Each question names a reduced model inside the full one, and the natural measure of what the full model adds is how much it lowers the residual sum of squares. This section develops that measure, its computation when regressors are added to a fitted model, and a closed form when the reduced model is defined by linear constraints.

9.2.1 Notation

Definition 9.2.1. Extra sum of squares

Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be matrices with n rows, let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, and let $\mathbf{M}_1$ and $\mathbf{M}$ be the projections onto $\mathcal{C}(\mathbf{X}_1)$ and $\mathcal{C}(\mathbf{X})$. The **extra sum of squares** of $\mathbf{X}_2$ given $\mathbf{X}_1$ is

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_1) \mathbf{y} = \text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X}),$$

with $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_1)$ degrees of freedom. Here $\text{SSE}(\cdot)$ is the residual sum of squares of the model with the given model matrix.

The two expressions agree by Eq. 6.5.2. One also writes $\text{SS}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)$, or $\text{SS}(B \mid \mu, A)$ in factor models; some books write R , for “reduction”. Four properties follow from Theorem 6.5.1 and Theorem 6.7.1.

1. *It depends only on the two column spaces.* Rescaling, recoding, centring, replacing $\mathbf{X}_2$ by $\mathbf{X}_2 + \mathbf{X}_1 \mathbf{B}$, and rank deficiency change nothing.
2. *It is nonnegative*, and zero iff $\mathbf{M} \mathbf{y} = \mathbf{M}_1 \mathbf{y}$, that is, iff adding $\mathbf{X}_2$ leaves the fitted values unchanged.
3. *It obeys a chain rule:* $\text{SS}(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = \text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) + \text{SS}(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$, because the projections telescope.
4. *It is the sum of squares of a residualized regression.* The next lemma makes this precise.

Lemma 9.2.1. Extra sums of squares by residualizing

For any $\mathbf{X}_1$ and $\mathbf{X}_2$, let $\tilde{\mathbf{X}}_2 = (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2$. Then $\mathcal{C}([\mathbf{X}_1, \mathbf{X}_2]) = \mathcal{C}(\mathbf{X}_1) \oplus \mathcal{C}(\tilde{\mathbf{X}}_2)$ with the two summands orthogonal, so $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$ and

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \|\mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)} \mathbf{y}\|^2 = \mathbf{y}^\top \tilde{\mathbf{X}}_2 (\tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}_2^\top \mathbf{y}.$$

Proof. The argument for part (a) of Lemma 6.6.1 uses no rank condition, so it applies here: the columns of $\tilde{\mathbf{X}}_2$ lie in $\mathcal{C}(\mathbf{X})$ and are orthogonal to $\mathcal{C}(\mathbf{X}_1)$, and every $\mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}_2 \mathbf{b}_2$ equals $\tilde{\mathbf{X}}_2 \mathbf{b}_2 + (\mathbf{X}_1 \mathbf{b}_1 + \mathbf{M}_1 \mathbf{X}_2 \mathbf{b}_2)$. By Theorem 6.3.7, $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$. The formula is Theorem 6.3.6 applied to $\tilde{\mathbf{X}}_2$. ■

9.2.2 Adding regressors to a fitted model

Suppose $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ has been fitted and we add the columns of an $n \times t$ matrix $\mathbf{Z}$, giving $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}$ with model matrix $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$. Only a $t \times t$ matrix has to be inverted.

Theorem 9.2.2. Adding regressors

Let $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$ have full column rank $p + t$, let $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$, $\hat{\boldsymbol{\beta}} = \mathbf{C}\mathbf{X}^\top \mathbf{y}$, and $\mathbf{R} = \mathbf{I} - \mathbf{X}\mathbf{C}\mathbf{X}^\top$. Then:

- (a) $\mathbf{Z}^\top \mathbf{R}\mathbf{Z}$ is positive definite;
- (b) the least squares coefficients of the enlarged model are

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R}\mathbf{y}, \quad \hat{\boldsymbol{\beta}}_W = \hat{\boldsymbol{\beta}} - \mathbf{L}\hat{\boldsymbol{\gamma}}, \quad \mathbf{L} = \mathbf{C}\mathbf{X}^\top \mathbf{Z};$$

- (c) the residual sum of squares falls by

$$\text{SS}(\mathbf{Z} \mid \mathbf{X}) = \mathbf{y}^\top \mathbf{R}\mathbf{Z}(\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R}\mathbf{y} = \hat{\boldsymbol{\gamma}}^\top \mathbf{Z}^\top \mathbf{R}\mathbf{y} = \hat{\boldsymbol{\gamma}}^\top (\mathbf{Z}^\top \mathbf{R}\mathbf{Z}) \hat{\boldsymbol{\gamma}};$$

- (d) if $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then $\text{Cov}(\hat{\boldsymbol{\gamma}}) = \sigma^2 (\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1}$, $\text{Cov}(\hat{\boldsymbol{\beta}}_W, \hat{\boldsymbol{\gamma}}) = -\mathbf{L} \text{Cov}(\hat{\boldsymbol{\gamma}})$ and

$$\text{Cov}(\hat{\boldsymbol{\beta}}_W) = \sigma^2 [\mathbf{C} + \mathbf{L}(\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} \mathbf{L}^\top].$$

Proof.

- (a) $\mathbf{R}$ is the symmetric idempotent residual projection of the model $\mathbf{X}$, so $\mathbf{a}^\top \mathbf{Z}^\top \mathbf{R}\mathbf{Z}\mathbf{a} = \|\mathbf{R}\mathbf{Z}\mathbf{a}\|^2 \geq 0$. If $\mathbf{R}\mathbf{Z}\mathbf{a} = \mathbf{0}$, then $\mathbf{Z}\mathbf{a} \in \mathcal{C}(\mathbf{X})$, say $\mathbf{Z}\mathbf{a} = \mathbf{X}\mathbf{b}$, so $\mathbf{W}(-\mathbf{b}^\top, \mathbf{a}^\top)^\top = \mathbf{0}$ and full column rank forces $\mathbf{a} = \mathbf{0}$.
- (b) Apply the Frisch–Waugh–Lovell theorem (Theorem 6.6.2) with $\mathbf{X}_1 = \mathbf{Z}$ and $\mathbf{X}_2 = \mathbf{X}$. Then $\mathbf{M}_2 = \mathbf{I} - \mathbf{R}$ and $\tilde{\mathbf{X}}_1 = \mathbf{R}\mathbf{Z}$, so part (a) of that theorem gives $\hat{\boldsymbol{\gamma}} = (\mathbf{Z}^\top \mathbf{R}\mathbf{R}\mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R}\mathbf{y}$, which is the stated formula because $\mathbf{R}$ is symmetric and idempotent. Part (c) of the theorem gives $\hat{\boldsymbol{\beta}}_W = \mathbf{C}\mathbf{X}^\top (\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}) = \hat{\boldsymbol{\beta}} - \mathbf{L}\hat{\boldsymbol{\gamma}}$.
- (c) By Lemma 9.2.1, the extra sum of squares is the squared length of the projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{R}\mathbf{Z})$, which is $\mathbf{y}^\top \mathbf{R}\mathbf{Z}(\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R}\mathbf{y}$ by Theorem 6.3.6. Substituting (b) gives the other two forms.
- (d) Both $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\beta}}$ are linear in $\mathbf{y}$, and by Theorem 2.2.1

$$\text{Cov}(\hat{\boldsymbol{\gamma}}) = \sigma^2 (\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R}\mathbf{R}\mathbf{Z}(\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} = \sigma^2 (\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}}) = \sigma^2 \mathbf{C}\mathbf{X}^\top \mathbf{R}\mathbf{Z}(\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} =$$

since $\mathbf{X}^\top \mathbf{R} = \mathbf{0}$. With $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{C}$ (Theorem 5.5.1), the formula $\hat{\boldsymbol{\beta}}_W = \hat{\boldsymbol{\beta}} - \mathbf{L}\hat{\boldsymbol{\gamma}}$ gives $\text{Cov}(\hat{\boldsymbol{\beta}}_W, \hat{\boldsymbol{\gamma}}) = -\mathbf{L} \text{Cov}(\hat{\boldsymbol{\gamma}})$ and $\text{Cov}(\hat{\boldsymbol{\beta}}_W) = \text{Cov}(\hat{\boldsymbol{\beta}}) + \mathbf{L} \text{Cov}(\hat{\boldsymbol{\gamma}}) \mathbf{L}^\top$. ■

Since $\mathbf{L}(\mathbf{Z}^\top \mathbf{R}\mathbf{Z})^{-1} \mathbf{L}^\top$ is nonnegative definite, part (d) shows that *adding regressors never decreases the variances of the coefficients already in the model*, when both models are correct (compare Exercise (B2), and see Exercise (C2)). Chapter 26 measures this price of adjustment as variance inflation (Definition 26.2.1). In computation one updates a QR factorization instead of using these formulas (Chapter 10).

Example. Adding poverty and urbanization

In the state data, fit the murder rate on single parenthood first: $\hat{y} = -8.2146 + 0.5147 x_{\text{single}}$, with residual sum of squares 121.92. Now add $\mathbf{Z} = [\mathbf{x}_{\text{poverty}}, \mathbf{x}_{\text{urban}}]$. The listing computes $\hat{\gamma} = (0.2538, 0.0046)$ from the old residual projection alone, and corrects the old coefficients to -9.0946 and 0.3980 . These agree with a fresh fit of the enlarged model. The residual sum of squares falls by 19.96, on 2 degrees of freedom, to 101.96. Single parenthood's coefficient drops by almost a quarter once poverty is allowed to take its share, the pattern seen in Example (Order matters).

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}

X = np.column_stack([one, cols["single"]])           # the model already
    fitted
Z = np.column_stack([cols["poverty"], cols["urban"]]) # the regressors to add
XtX_inv = np.linalg.inv(X.T @ X)
beta_X = XtX_inv @ X.T @ y
R = np.eye(n) - X @ XtX_inv @ X.T                  # residual projection of
    the old model

gamma = np.linalg.solve(Z.T @ R @ Z, Z.T @ R @ y)   # new coefficients
beta_new = beta_X - XtX_inv @ X.T @ Z @ gamma       # old coefficients,
    corrected
extra_ss = gamma @ Z.T @ R @ y                      # SS(Z | X) = SSE_X -
    SSE_W
print("gamma =", gamma.round(4), " corrected beta =", beta_new.round(4))
print(f"SSE_X = {y @ R @ y:.2f}, extra SS = {extra_ss:.2f}")
```

[Run this code online](#)

9.2.3 One column at a time

With $t = 1$, [Theorem 9.2.2](#) links the extra sum of squares of a single regressor to its coefficient and standard error.

Corollary 9.2.3. The extra sum of squares of one column

Let $\mathbf{X}$ have full column rank p , write c_{jj} for the j th diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$, and let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ without its j th column. Then

$$\text{SS}(\mathbf{x}_j | \mathbf{X}_{(j)}) = \frac{\hat{\beta}_j^2}{c_{jj}} = t_j^2 s^2, \quad t_j = \frac{\hat{\beta}_j}{s\sqrt{c_{jj}}}, \quad s^2 = \frac{\text{SSE}}{n - p}.$$

Consequently, the ratio of this one-degree-of-freedom sum of squares to the residual mean square is t_j^2 .

Proof. Permute the columns so that $\mathbf{x}_j$ comes last; this carries c_{jj} to the last diagonal entry of the inverse. Write $\mathbf{X} = [\mathbf{X}_{(j)}, \mathbf{x}_j]$ and $\mathbf{R} = \mathbf{I} - \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top$ for the residual projection of $\mathbf{X}_{(j)}$. By Theorem 1.6.4 the last diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$ is the inverse of the Schur complement $\mathbf{x}_j^\top \mathbf{x}_j - \mathbf{x}_j^\top \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top \mathbf{x}_j = \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j$, so $\mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = 1/c_{jj}$. Now apply Theorem 9.2.2 with $\mathbf{X}_{(j)}$ as the old model and $\mathbf{Z} = \mathbf{x}_j$: then $\hat{\gamma} = \hat{\beta}_j$, and part (c) gives $\text{SS}(\mathbf{x}_j \mid \mathbf{X}_{(j)}) = \hat{\beta}_j^2 \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = \hat{\beta}_j^2 / c_{jj}$. The definition of t_j gives the last form. ■

For poverty in the three-regressor model, $\hat{\beta} = 0.2538$ and $c_{jj} = 0.003935$, so the extra sum of squares is $0.2538^2 / 0.003935 = 16.37$. This is exactly the last entry for poverty in Order B of Example (Order matters). The t statistic is 2.718, and $t^2 s^2 = 7.386 \times 2.217 = 16.37$.

```
Xf = np.column_stack([one, cols["poverty"], cols["single"], cols["urban"]])
fit = sm.OLS(y, Xf).fit()
C = np.linalg.inv(Xf.T @ Xf)
j = 1 # poverty
partial_ss = fit.params[j] ** 2 / C[j, j]
print(f"beta = {fit.params[j]:.4f}, c_jj = {C[j, j]:.6f}, beta^2/c_jj = {partial_ss:.2f}")
print(f"t = {fit.tvalues[j]:.3f}, t^2 * s^2 = {fit.tvalues[j] ** 2 * fit.scale:.2f}")
```

[Run this code online](#)

9.2.4 Sums of squares for linear hypotheses

Reduced models often come as constraints: two coefficients are equal, a set of coefficients vanishes, a contrast among group means is zero. Section 6.5 identified the test space of such a constraint (Theorem 6.5.2). The next theorem gives its sum of squares in terms of the least squares estimate, with no reduced model fitted.

Theorem 9.2.4. Sum of squares of a linear hypothesis

Let $\mathbf{\Lambda}$ be a $p \times q$ matrix of rank q such that each entry of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable, so that $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ for some $n \times q$ matrix $\mathbf{T} = [\boldsymbol{\rho}_1, \dots, \boldsymbol{\rho}_q]$ (Definition 8.2.1, Theorem 6.4.4). Let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$, let $\hat{\boldsymbol{\beta}}$ be any least squares estimate, and let $\mathbf{d} \in \mathbb{R}^q$. Then:

- (a) $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ do not depend on the choices of $\hat{\boldsymbol{\beta}}$ and $\mathbf{G}$, and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is positive definite;
- (b) the subspace $\mathcal{S}_0 = \{\mathbf{X} \mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ of $\mathcal{C}(\mathbf{X})$ has orthogonal complement in $\mathcal{C}(\mathbf{X})$ equal to $\mathcal{C}(\mathbf{M} \mathbf{T})$, of dimension q , and the projection onto $\mathcal{C}(\mathbf{M} \mathbf{T})$ is $\mathbf{M} \mathbf{T} (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M}$;
- (c) the minimum of $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2$ over $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$ exceeds SSE by

$$\text{SS}_H = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}). \quad (9.2.1)$$

Proof.

- (a) $\Lambda^\top \hat{\beta} = \mathbf{T}^\top \mathbf{X} \hat{\beta} = \mathbf{T}^\top \mathbf{M} \mathbf{y}$ and $\Lambda^\top \mathbf{G} \Lambda = \mathbf{T}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{T}^\top \mathbf{M} \mathbf{T}$ by Theorem 6.3.6, and neither expression involves the choices. Since $\mathbf{X}^\top \mathbf{M} = \mathbf{X}^\top$, we have $\mathbf{X}^\top (\mathbf{M} \mathbf{T}) = \Lambda$, which has rank q , so the q columns of $\mathbf{M} \mathbf{T}$ are linearly independent. Hence $\mathbf{T}^\top \mathbf{M} \mathbf{T} = (\mathbf{M} \mathbf{T})^\top (\mathbf{M} \mathbf{T})$ is positive definite.
- (b) $\Lambda^\top \mathbf{b} = \mathbf{T}^\top \mathbf{X} \mathbf{b}$, so $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}^\top)$. Theorem 6.5.2 identifies its orthogonal complement in $\mathcal{C}(\mathbf{X})$ as $\mathcal{C}(\mathbf{M} \mathbf{T})$. The columns of $\mathbf{M} \mathbf{T}$ are independent, so Theorem 6.3.6 gives the projection $\mathbf{M} \mathbf{T} (\mathbf{T}^\top \mathbf{M} \mathbf{T})^{-1} \mathbf{T}^\top \mathbf{M}$, and $\mathbf{T}^\top \mathbf{M} \mathbf{T} = \Lambda^\top \mathbf{G} \Lambda$ by (a).
- (c) First let $\mathbf{d} = \mathbf{0}$. Minimizing over $\Lambda^\top \mathbf{b} = \mathbf{0}$ is minimizing $\|\mathbf{y} - \mu\|^2$ over $\mu \in \mathcal{S}_0$, so by Eq. 6.5.2 the excess over SSE is the squared length of the projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0 = \mathcal{C}(\mathbf{M} \mathbf{T})$, where $\ominus$ denotes the orthogonal complement within $\mathcal{C}(\mathbf{X})$. By (b) this is $\mathbf{y}^\top \mathbf{M} \mathbf{T} (\Lambda^\top \mathbf{G} \Lambda)^{-1} \mathbf{T}^\top \mathbf{M} \mathbf{y}$, and $\mathbf{T}^\top \mathbf{M} \mathbf{y} = \Lambda^\top \hat{\beta}$. For general $\mathbf{d}$, the rows of $\Lambda^\top$ are independent, so there is a $\mathbf{b}_0$ with $\Lambda^\top \mathbf{b}_0 = \mathbf{d}$. Substituting $\mathbf{b} = \mathbf{b}_0 + \mathbf{b}_*$ turns the problem into minimizing $\|\mathbf{y}_* - \mathbf{X} \mathbf{b}_*\|^2$ subject to $\Lambda^\top \mathbf{b}_* = \mathbf{0}$, with $\mathbf{y}_* = \mathbf{y} - \mathbf{X} \mathbf{b}_0$. The data $\mathbf{y}_*$ have the same residual sum of squares as $\mathbf{y}$, and $\hat{\beta} - \mathbf{b}_0$ is a least squares estimate for them, since $\mathbf{X}(\hat{\beta} - \mathbf{b}_0) = \mathbf{M} \mathbf{y}_*$. Applying the case $\mathbf{d} = \mathbf{0}$ to $\mathbf{y}_*$ gives Eq. 9.2.1, because $\Lambda^\top (\hat{\beta} - \mathbf{b}_0) = \Lambda^\top \hat{\beta} - \mathbf{d}$.

■

In Eq. 9.2.1, $\Lambda^\top \hat{\beta} - \mathbf{d}$ measures how far the estimate is from the hypothesis, and $\sigma^2 \Lambda^\top \mathbf{G} \Lambda$ is the covariance of $\Lambda^\top \hat{\beta}$ (Chapter 7), so SS_H / σ^2 is the squared Mahalanobis length of the discrepancy (Proposition 3.2.3). No reduced model has to be fitted.

In the full-rank case the constrained minimizer itself has a closed form, which recovers the classical restricted least squares estimator.

Theorem 9.2.5. Restricted least squares

Let $\mathbf{X}$ have full column rank, $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$, and let Λ ($p \times q$) have rank q . For any $\mathbf{d} \in \mathbb{R}^q$, the unique minimizer of $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2$ subject to $\Lambda^\top \mathbf{b} = \mathbf{d}$ is

$$\hat{\beta}_H = \hat{\beta} - \mathbf{C} \Lambda (\Lambda^\top \mathbf{C} \Lambda)^{-1} (\Lambda^\top \hat{\beta} - \mathbf{d}),$$

and $\|\mathbf{y} - \mathbf{X} \hat{\beta}_H\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\beta} - \hat{\beta}_H)\|^2$, where the last term equals SS_H of Eq. 9.2.1 with $\mathbf{G} = \mathbf{C}$.

Proof. $\Lambda^\top \mathbf{C} \Lambda$ is positive definite because $\mathbf{C}$ is and Λ has full column rank, and $\Lambda^\top \hat{\beta}_H = \Lambda^\top \hat{\beta} - (\Lambda^\top \hat{\beta} - \mathbf{d}) = \mathbf{d}$. For any $\mathbf{b}$, Eq. 6.3.1 with $\mathbf{u} = \mathbf{X} \mathbf{b}$ gives $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\beta} - \mathbf{b})\|^2$. Every feasible $\mathbf{b}$ has the form $\hat{\beta}_H + \mathbf{v}$ with $\Lambda^\top \mathbf{v} = \mathbf{0}$, and then

$$\|\mathbf{X}(\hat{\beta} - \mathbf{b})\|^2 = \|\mathbf{X}(\hat{\beta} - \hat{\beta}_H)\|^2 - 2(\hat{\beta} - \hat{\beta}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \|\mathbf{X} \mathbf{v}\|^2.$$

The cross term vanishes: $(\hat{\beta} - \hat{\beta}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\Lambda^\top \hat{\beta} - \mathbf{d})^\top (\Lambda^\top \mathbf{C} \Lambda)^{-1} \Lambda^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\Lambda^\top \hat{\beta} - \mathbf{d})^\top (\Lambda^\top \mathbf{C} \Lambda)^{-1} \Lambda^\top \mathbf{v} = 0$. So the objective is $\text{SSE} + \|\mathbf{X}(\hat{\beta} - \hat{\beta}_H)\|^2 + \|\mathbf{X} \mathbf{v}\|^2$, minimized exactly when $\mathbf{X} \mathbf{v} = \mathbf{0}$, that is, $\mathbf{v} = \mathbf{0}$. Finally $\|\mathbf{X}(\hat{\beta} - \hat{\beta}_H)\|^2 = (\Lambda^\top \hat{\beta} - \mathbf{d})^\top (\Lambda^\top \mathbf{C} \Lambda)^{-1} \Lambda^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{C} \Lambda (\Lambda^\top \mathbf{C} \Lambda)^{-1} (\Lambda^\top \hat{\beta} - \mathbf{d})$, which simplifies to SS_H .

■

The same estimator comes out of Lagrange multipliers (Proposition 1.11.3 with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$).

The proof above shows instead that the constrained fit is the feasible point nearest to $\hat{\beta}$ in the geometry of $\mathbf{X}^\top \mathbf{X}$.

Example. Equal effects of poverty and single parenthood

Both poverty and single parenthood are percentages of the population. Is one percentage point of either associated with the same change in the murder rate? The hypothesis is $\lambda^\top \beta = 0$ with $\lambda = (0, 1, -1, 0)^\top$. Formula Eq. 9.2.1 gives $SS_H = 1.807$ on one degree of freedom, identical to the drop in residual sum of squares when the two regressors are replaced by their sum. Its ratio to the residual mean square is 0.815, so the data are entirely compatible with equal effects. For a hypothesis with a nonzero right-hand side, $\beta_{\text{single}} = 0.5$, Theorem 9.2.5 gives the constrained coefficients 0.1803 for poverty and -0.0044 for urbanization, and a sum of squares of 3.357, both confirmed by a direct fit of the reduced model.

```
def hypothesis_ss(X, y, Lam, d=None):
    """(L'b - d)' [L' (X'X) ~ L] ~{-1} (L'b - d) for estimable L'beta = d."""
    G = np.linalg.pinv(X.T @ X)
    b = G @ X.T @ y
    u = Lam.T @ b - (0 if d is None else d)
    return u @ np.linalg.solve(Lam.T @ G @ Lam, u)

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

Lam_equal = np.array([[0.0, 1.0, -1.0, 0.0]]).T          # beta_poverty =
beta_single
ss_equal = hypothesis_ss(Xf, y, Lam_equal)
X_equal = np.column_stack([one, cols["poverty"] + cols["single"], cols["urban"]])
print(f"H: equal slopes    formula {ss_equal:.3f}    SSE_0 - SSE {sse(X_equal) - sse(
Xf):.3f}")
```

[Run this code online](#)

9.2.5 Constraints that are not estimable

Without estimability, the reduced model $\mathcal{S}_0 = \{\mathbf{Xb} : \Lambda^\top \mathbf{b} = \mathbf{0}\}$ is still defined, but it may be larger than the number of constraints suggests.

Proposition 9.2.6. How much a constraint restricts

For any $p \times q$ matrix Λ , the subspace $\mathcal{S}_0 = \{\mathbf{Xb} : \Lambda^\top \mathbf{b} = \mathbf{0}\}$ has dimension $r - \dim(\mathcal{C}(\Lambda) \cap \mathcal{C}(\mathbf{X}^\top))$. So the extra sum of squares for the constraint has $\dim(\mathcal{C}(\Lambda) \cap \mathcal{C}(\mathbf{X}^\top))$ degrees of freedom. It is identically zero iff no nonzero linear combination of the constrained functions is estimable.

Proof. $\mathcal{S}_0$ is the image of $\mathcal{N}(\Lambda^\top)$ under $\mathbf{b} \mapsto \mathbf{Xb}$, and the kernel of this map restricted to $\mathcal{N}(\Lambda^\top)$ is $\mathcal{N}(\Lambda^\top) \cap \mathcal{N}(\mathbf{X})$, the null space of the stacked matrix $[\Lambda, \mathbf{X}^\top]^\top$. By rank-nullity

(Theorem 1.2.2),

$$\dim \mathcal{S}_0 = \dim \mathcal{N}(\mathbf{\Lambda}^\top) - \dim(\mathcal{N}(\mathbf{\Lambda}^\top) \cap \mathcal{N}(\mathbf{X})) = (p - \text{rank } \mathbf{\Lambda}) - (p - \text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top]).$$

The dimension formula for a sum of subspaces (Section 6.2) gives $\text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top] = \dim(\mathcal{C}(\mathbf{\Lambda}) + \mathcal{C}(\mathbf{X}^\top)) = \text{rank } \mathbf{\Lambda} + r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. Substituting, $\dim \mathcal{S}_0 = r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. The degrees of freedom of the extra sum of squares are $r - \dim \mathcal{S}_0$. A combination $\mathbf{\Lambda}\mathbf{a}$ is estimable iff it lies in $\mathcal{C}(\mathbf{X}^\top)$ (Theorem 6.4.4), which gives the last statement. ■

In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$, the constraint $\alpha_1 = \alpha_2$ has one degree of freedom. The constraint $\alpha_1 = 0$ has none: it does not restrict the mean vector, and only picks one of the least squares solutions (Theorem 8.4.2). Software asked to “test” a nonestimable hypothesis either refuses or silently tests its estimable part. Chapter 17 meets this with empty cells (Theorem 17.4.1).

9.2.6 Exercises

9.2.6.1 A. Check your understanding

Exercise. A1

In simple regression, compute SS_H for $\beta_1 = 0$ from Eq. 9.2.1 and show that it equals $\hat{\beta}_1^2 S_{xx}$, in agreement with Exercise (A1).

9.2.6.2 B. Practice

Exercise. B1

Prove the chain rule $SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = SS(\mathbf{X}_2 \mid \mathbf{X}_1) + SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$, and deduce $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1)$. Show by an example from Example (Adding poverty and urbanization) or its listing that $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_3 \mid \mathbf{X}_1)$ can fail.

Solution. Let $\mathbf{M}_1$, $\mathbf{M}_{12}$ and $\mathbf{M}_{123}$ be the projections onto the nested column spaces. Then $\mathbf{M}_{123} - \mathbf{M}_1 = (\mathbf{M}_{12} - \mathbf{M}_1) + (\mathbf{M}_{123} - \mathbf{M}_{12})$, and the chain rule follows by taking $\mathbf{y}^\top(\cdot)\mathbf{y}$. Both terms are nonnegative, which gives the inequality. For the failure, the listing of Section 9.3 reports $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}) = 6.30$ but $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}, \mathbf{x}_{\text{poverty}}) = 29.26$. Adjusting for poverty more than quadruples the sum of squares for urbanization.

Exercise. B2

In Theorem 9.2.5 with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, show that

$$\text{Cov}(\hat{\boldsymbol{\beta}}_H) = \sigma^2 [\mathbf{C} - \mathbf{C}\mathbf{\Lambda}(\mathbf{\Lambda}^\top \mathbf{C}\mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C}],$$

that $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_H)$ is nonnegative definite, and that $E(\hat{\boldsymbol{\beta}}_H) = \boldsymbol{\beta}$ when the constraint is true. What is $E(\hat{\boldsymbol{\beta}}_H)$ when it is false?

Solution. Write $\mathbf{K} = \mathbf{C}\mathbf{A}(\mathbf{A}^\top\mathbf{C}\mathbf{A})^{-1}$, so $\hat{\boldsymbol{\beta}}_H = (\mathbf{I} - \mathbf{K}\mathbf{A}^\top)\hat{\boldsymbol{\beta}} + \mathbf{K}\mathbf{d}$. By Theorem 2.2.1 and $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{C}$, $\text{Cov}(\hat{\boldsymbol{\beta}}_H) = \sigma^2(\mathbf{I} - \mathbf{K}\mathbf{A}^\top)\mathbf{C}(\mathbf{I} - \mathbf{A}\mathbf{K}^\top)$. Expanding and using $\mathbf{A}^\top\mathbf{C}\mathbf{A}(\mathbf{A}^\top\mathbf{C}\mathbf{A})^{-1} = \mathbf{I}$, the two cross terms and the last term all equal $\mathbf{K}\mathbf{A}^\top\mathbf{C} = \mathbf{C}\mathbf{A}(\mathbf{A}^\top\mathbf{C}\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{C}$, with signs $-, -, +$, which gives the formula. The difference $\sigma^2\mathbf{C}\mathbf{A}(\mathbf{A}^\top\mathbf{C}\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{C}$ is of the form $\mathbf{B}\mathbf{A}\mathbf{B}^\top$ with $\mathbf{A}$ positive definite, hence nonnegative definite. Finally $\mathbf{E}(\hat{\boldsymbol{\beta}}_H) = \boldsymbol{\beta} - \mathbf{K}(\mathbf{A}^\top\boldsymbol{\beta} - \mathbf{d})$, which is $\boldsymbol{\beta}$ iff $\mathbf{A}^\top\boldsymbol{\beta} = \mathbf{d}$. A false constraint buys the smaller variance at the price of a bias.

Exercise. B3

In the one-way layout with model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$ and group sizes $n_1, \dots, n_g$, use the generalized inverse $\mathbf{G} = \text{diag}(0, 1/n_1, \dots, 1/n_g)$ in Eq. 9.2.1 to show that the sum of squares for $\alpha_1 = \alpha_2$ is $(\bar{y}_1 - \bar{y}_2)^2/(1/n_1 + 1/n_2)$. Verify with Proposition 9.2.6 that the constraint $\alpha_1 = 0$ has zero degrees of freedom.

Solution. $\mathbf{G}$ is a generalized inverse because $\mathbf{X}^\top\mathbf{X}$ has the block form with first row $(n, n_1, \dots, n_g)$ and lower right block $\text{diag}(n_1, \dots, n_g)$, and a direct multiplication gives $\mathbf{X}^\top\mathbf{X}\mathbf{G}\mathbf{X}^\top\mathbf{X} = \mathbf{X}^\top\mathbf{X}$. With $\boldsymbol{\lambda} = (0, 1, -1, 0, \dots, 0)^\top$, $\mathbf{G}\mathbf{X}^\top\mathbf{y} = (0, \bar{y}_1, \dots, \bar{y}_g)^\top$, so $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \bar{y}_1 - \bar{y}_2$ and $\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = 1/n_1 + 1/n_2$. For $\alpha_1 = 0$, $\boldsymbol{\lambda} = (0, 1, 0, \dots, 0)^\top$ is not in $\mathcal{C}(\mathbf{X}^\top)$ (Example (Four answers, one fit)), and $\mathcal{C}(\boldsymbol{\lambda})$ is a line, so $\mathcal{C}(\boldsymbol{\lambda}) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$.

9.2.6.3 C. Going deeper

Exercise. C1

Using Exercise (C1) and Lemma 9.2.1, show that

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \max_{\mathbf{a}} \frac{(\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \mathbf{y})^2}{\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2 \mathbf{a}},$$

the maximum being over $\mathbf{a}$ with $\tilde{\mathbf{X}}_2\mathbf{a} \neq \mathbf{0}$. Deduce that the extra sum of squares of a block is at least that of any single column of the block, given $\mathbf{X}_1$, and that the sum of squares of an estimable hypothesis $\mathbf{A}^\top\boldsymbol{\beta} = \mathbf{0}$ equals $\max_{\mathbf{a}} (\mathbf{a}^\top \mathbf{A}^\top \hat{\boldsymbol{\beta}})^2 / (\mathbf{a}^\top \mathbf{A}^\top \mathbf{G} \mathbf{A} \mathbf{a})$.

Exercise. C2

Exercise (B2) gives the matrix inequality behind the remark after Theorem 9.2.2. Using part (d) of that theorem instead, show that $\text{Var}(\hat{\beta}_{W,j}) = \text{Var}(\hat{\beta}_j)$ iff the j th row of $\mathbf{L}$ is zero, and that this holds for every j iff $\mathbf{X}^\top\mathbf{Z} = \mathbf{0}$. Show that when $\mathbf{X}^\top\mathbf{Z} = \mathbf{0}$ the estimates $\hat{\boldsymbol{\beta}}$ do not change at all.

Exercise. C3

In the additive two-way model $\mathbf{E}(Y_{ijk}) = \mu + \alpha_i + \beta_j$ with all cells occupied, use Proposi-

tion 9.2.6 to find the degrees of freedom of the extra sums of squares for the constraints (i) $\sum_i \alpha_i = 0$ and $\sum_j \beta_j = 0$; (ii) $\alpha_1 = \alpha_2$; (iii) $\alpha_1 = 0$ and $\alpha_1 = \alpha_2$.